Motor Imagery–Based Brain–Computer Interfaces: Challenges, Methods, and Future Directions

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***Abstract*—Brain–computer interfaces (BCIs) offer direct interaction between human brain and external devices, bypassing peripheral nerves and muscles. Within noninvasive BCI, electroencephalography (EEG) stands out due to its affordability and high temporal resolution. Motor imagery (MI)–based BCIs, in particular, harness the user's imagination of body movements to generate control commands without external triggers. This survey aims to synthesize recent literature on MI–based BCIs, providing an overview of key methods, existing challenges, and prospective solutions. By focusing on thematic insights, we highlight how researchers are refining signal preprocessing, feature extraction, and classification strategies to boost system performance. We conducted a thematic analysis of major works in EEG acquisition, artifact removal, feature extraction methods, classification approaches,deep neural networks, and adaptive or transfer learning frameworks. We also explored user-centered factors such as training protocols and the phenomenon of BCI–illiteracy. Notable advancements have emerged in signal preprocessing, sophisticated machine learning models, and calibration-reducing strategies like transfer learning. However, the complexity of EEG signals, high inter-subject variability, and the presence of BCI–illiteracy still limit real-world adoption. Adaptive techniques and personalized feedback can mitigate these constraints, and novel methods continue to emerge. MI–BCIs hold promise for rehabilitation, assistive technologies, neuroergonomics, and broader human–machine interactions. By clarifying these core themes this survey underscores the need for multidisciplinary, user-centric approaches to advance MI–BCI reliability, accuracy, and overall usability in day-to-day contexts.**

***Index Terms*—Brain-Computer Interface; Motor Imagery; EEG; Transfer Learning; Adaptive Decoding, Deep Learning**

I. Introduction

Brain–computer interfaces (BCI), AKA brain–machine interface (BMI), facilitate a direct communication pathway between neural activity and external devices, effectively translating thought processes into actions [1]–[5]. It requires no activation of muscles or peripheral nerve. The past four decades have witnessed sophisticated BCI–controlled applications [6], and tons of related literature [4].

Among various BCI modalities like magnetoencephalography (MEG) and functional magnetic resonance imaging (fMRI), electroencephalography (EEG) is especially popular thanks to its noninvasiveness, relative portability, convenience, and high temporal resolution [7]–[9]. Sensors captures the electromagnetic activity of the brain containing many biomarkers which can be exploited to reflect on the status of the brain and its activity [3], [4]. Decoding of EEG signal is one of the most important enablers of BCI technology [10].

Within EEG–based BCIs, motor imagery (MI) plays a central role. MI allows users to imagine specific movements—such as clenching a fist or flexing a wrist—to generate sensorimotor rhythms (SMRs) in the α (8–13 Hz) and β (13–30 Hz) frequency bands [2], [11]. Because the user internally initiates these movements, MI–BCIs support asynchronous control, enabling interactions without dependence on external cues such as flashes or beeps [2].

Despite substantial research strides, a range of hurdles continues to hamper practical deployments. Variability of EEG patterns, BCI–illiteracy, and the computational overhead of real-time signal decoding all constrain MI-BCI adoption [12]–[14]. Ongoing work focuses on advanced machine learning methods, adaptive calibration techniques, and user training frameworks to meet the demands of real-world BCI applications.

In compiling this survey, we adopted a thematic analysis to organize the substantial volume of literature on MI–BCI. Thematic analysis helps cluster related concepts into coherent categories, allowing a multifaceted understanding of the technology and its challenges. Specifically, we examined articles spanning signal acquisition and processing, feature extraction, classification techniques, phenomenon of BCI–illiteracy, and user training or adaptation strategies. We then consolidated these into themes:

- Foundational Aspects of MI–BCI: Overview of EEG signals and their salient features for MI–based applications.
- Core Technical Modules: Preprocessing, artifact removal, feature extraction, and machine learning classifiers.
- Challenges and Limitations: Low SNR, variability in MI–EEG, data scarcity, and BCI–illiteracy.
- Current Research: Methods that mitigate nonstationary signals, reduce calibration burdens, and enhance generalization.
- Advancements and Future Directions: Strategies for improving user engagement, and cutting-edge research to push MI–BCI toward broader adoption.

By mapping the field according to these themes, our paper underscores the interplay between technology innovations and user-driven constraints, capturing progress while spotlighting areas still in need of exploration.

II. Foundations of Motor Imagery–Based EEG

A. EEG Signal Characteristics

EEG records electrical brain activity by placing electrodes on the scalp according to standardized configurations (e.g., 10–10 system). Its primary virtue lies in high temporal resolution, which enables tracking rapid changes in cortical activity in real time [9]. Yet EEG signals inherently suffer from low SNR and are influenced by noise sources including muscle artifacts (EMG), eye-blink artifacts (EOG), and ambient electromagnetic interference [4], [8]. These confounding factors necessitate rigorous preprocessing measures—such as filtering, artifact rejection, and sometimes advanced decomposition algorithms. The EEG signal is a time series [8] which can be expressed as a matrix of data samples over time for all channels resulted from the different electrodes. Yet, EEG signals are characteristically low in signal–to–noise ratio (SNR) and susceptible to physiological and environmental artifacts, presenting researchers with a multitude of challenges [4]. Also, EEG suffers from low spatial resolution [15] where the electrodes detect the electromagnetic activity from allover the scalp.

B. Motor Imagery

BCI paradigms can be categorized by 1) the mode of operation and 2) the degree of reliance on external stimuli [4], [12], [16]. In synchronized BCIs, the user's control command is triggered at a system-defined time, typically presented via cue or stimulus. In contrast, asynchronous BCIs allow users to generate control signals spontaneously at any moment, with the BCI detecting and interpreting these signals as they emerge. A parallel classification considers 1) evoked potentials versus 2) spontaneous (also termed "voluntary") signals. If the brain signal is generated in response to an external stimulus, it is classified as evoked-potential; if it arises from the user's freewill without external triggering, it is considered spontaneous [12], [17]; these are sometimes referred to as active and passive BCI systems, respectively [16].

MI–BCI paradigms offer asynchronous operation and spontaneity, in contrast to evoked-potential BCIs (like P300 or SSVEP) that rely on external stimuli [2], [4] . This advantage makes MI–BCIs more intuitive but also more subject to inter-subject and intra-subject variability [18]. The success of MI–BCI thus hinges on reliable detection of subtle neural correlates of imagined actions in highly variable EEG signals.

C. Sensorimotor Rhythms

Motor imagery relies on the cognitive act of simulating movement without performing it physically [2], [4], [19]–[21]. When a user imagines moving a limb, the sensorimotor cortex exhibits event–related desynchronization (ERD) and event–related synchronization (ERS) in the α (often labeled μ in sensorimotor areas) and β frequency ranges [2], [22]. These oscillatory patterns, collectively known as SMRs, serve as the fundamental basis for MI–BCI classification. The MI–BCI module recognizes those oscillatory patterns and generates control commands to the devices/application/machine accordingly [23]. For instance, imagining flexing the left wrist typically results in a characteristic ERD/ERS localized around the corresponding cortical representation [3], [21].

III. Core Technical Modules in MI–BCI

A. Preprocessing and Artifact Removal

Preprocessing is a critical first step for in MI–BCI decoding. It enhances the SNR by mitigating artifacts before feature extraction. The surveyed literature has exhibited various techniques including filtering. Band-pass filters (e.g., Butterworth, elliptic) isolate α and β rhythms [2], [24] where the ERD/ERS exist. Moreover, removing artifacts like eye-blink (EOG) are tackled using independent component analysis (ICA), regression, or wavelet-based denoising, [25]. Spectral and Spatial Filtering was found in the literature as well, [26] combined Neural-Time-Series-Prediction-Preprocessing (NTSPP) with spectral filtering and Common Spatial Patterns (CSP) to enhance performance and reduce the number of required electrodes in multi-class MI–BCI. Combining different methods outperforms methods operating independently, reducing the number of electrodes required and significantly improving performance [26]. Hybrid feature extraction strategies, utilizing Multi-Scale Principal Component Analysis and deep learning architecture, achieved an impressive classification accuracy of around 99.2% [27].

B. Feature Extraction

Once the EEG is preprocessed, discriminative and informative features are required for effective decoding. Historically, Common Spatial Patterns (CSP) has been pivotal in MI–BCI, maximizing between-class variance by learning spatial filters [28]–[30]. Because its sensitivity to filter bands and time windows [31] its variations like Filter Bank Common Spatial Patterns (FBCSP) and Iterative Common Spatial-Temporal Patterns (ICSTP) are introduced in the literature. Filter Bank CSP (FBCSP) subdivides EEG into multiple frequency bands, applying CSP in each band to capture frequency-specific motor markers in the EEG [32]. On the other hand, Wavelet Transform and Time–Frequency Analyses provide fine-grained insights into EEG's nonstationary aspects [26], [27]. Some research have endeavors in reducing the number of electrodes. Selim et al. [33] proposed channel selection method with feature extraction using Root Mean Square (RMS) which resulted in better accuracy than state-of-the-art methods with a significant reduction in the number of channels. Hybrid and deep learning feature extraction is evident in contemporary works where combining CSP or wavelets with deep learning can lead to notable improvements. Stacked autoencoders or convolutional neural networks were used to model complex spatiotemporal patterns [27], [34].

C. Classification and Decoding

In classification, the classifier maps the discriminative features to specific classes associated with control commands. Many studies has emerged recently showing that deep learning have took a leading role in MI-BCI. Convolutional (CNNs), Recurrent (RNNs), and Transformer-based architectures can yield higher accuracy by learning intricate feature representations, though they typically need larger datasets and computational resources [35], [36]. See V-D for future direction on deep learning applications in MI-BCI. On the other hand, Linear Discriminant Analysis (LDA), Support Vector Machines (SVM), and Logistic Regression are widely employed due to low computational complexity [37]–[39].

Recent work on motor imagery (MI)–BCI systems has highlighted Riemannian geometry–based approaches as an effective way to handle EEG data as well. In Wang et al. [40] introduced a novel two-fold strategy. A sensitivity-based paradigm selection algorithm coupled with a generalized Riemann minimum distance mean for trial classification, leading to an average accuracy of 80.92% in binary classification. Riemannian metrics can address the inherent variability of EEG signals by measuring distances on covariance matrices. Additionally, various Riemannian-based metrics for MI–BCI performance using both simulated and real data were evaluated in [41]. They examined class separability and within-class consistency under different trial-wise adaptations—such as sliding window and weighted average approaches—demonstrating that careful application of Riemannian geometry can improve classifier robustness against nonstationary and subject-specific EEG patterns.

Alongside the approaches above, adaptive classification stands out for its ability to incrementally update model parameters to counteract the nonstationary nature of EEG signals [2], [11], [42]. By continuous training, adaptive classifiers can accommodate fluctuations introduced by fatigue, cognitive load, or changing mental states over a session [43], [44]. Researchers typically employ either supervised or unsupervised adaptations. Supervised approaches incorporate true class labels for each trial, whereas unsupervised methods adjust based on predicted labels alone [2], [42], [45]. While supervised adaptation can achieve higher accuracy, it requires continuous ground truth data, which might not be feasible in all real-time environments. Conversely, unsupervised adaptation is more autonomous but can risk classifier drift if initial predictions are inaccurate. Despite these trade-offs, adaptive classification remains an important strategy for dealing with EEG variability and improving MI–BCI reliability across different users and sessions.

IV. Challenges and Limitations

A. BCI–Illiteracy

Despite systematic training, a notable subset of users—estimated at 15–30%—cannot achieve consistent MI–BCI control [23], [46]. BCI–illiteracy, also called BCI inefficiency, can stem from various reasons. Neurophysiological Factors like anatomical differences or neuroanatomical irregularities might lead to undetectable SMRs [12], [47]. Inadequate mental imagery strategy or low engagement can degrade performance as well [48], [49]. Experimental design can affect the performance of the decoding as well. System parameters, electrode placements, or short training sessions may fail to capture the user's best MI patterns [50]. Nevertheless, no definitive reason of BCI-Illiteracy can be found in the literature. Moreover, endeavor is continuing to identify insufficient users and to overcome the BCI-illiteracy phenomena.

TABLE I
Advantages and Challenges of current research trends in MI–BCI

Technique	Advantages	Challenges
Adaptive Decoding	Improved performance [2], [53], [54] Enhanced user learning [55], [56]	Complexity [53], [57] Feedback dependency [58], [59]
Deep Learning	High accuracy [60], [61] Complex feature extraction [62], [63]	Data requirements [64], [65] Computational resources [65], [66]
Transfer Learning	Reduced calibration time [67]–[69] Improved performance [70]–[72]	Data alignment [70], [71] Variability [67]
Training & Feedback	Enhanced user engagement [58], [59], [73] Skill acquisition [73]–[75]	Feedback design [58], [59], [74] Variability in response [58], [73]

B. Inter–Subject and Intra–Subject Variability

EEG signals vary considerably across individuals due to differences in skull thickness, cortical geometry, and mental states [18]. Even within the same user, signals can drift over sessions, necessitating repeated calibration, [2], [11]. This lack of stability complicates the design of robust, real-time MI–BCI.

C. Data Insufficiency and Overfitting

EEG data collection labeling tasks is time-consuming and labor-intensive, often requiring repeated sessions for adequate coverage [11], [42]. Insufficient training samples can cause models—especially deep neural networks—to overfit and fail to generalize [10], [51].

D. Practical Usability Constraints

For daily usage, an MI–BCI system must be user-friendly, reliable, and responsive. Long calibration sessions, complex software setups can hinder adoption [33], [52]. Additionally, mental fatigue from sustained MI tasks affects throughput and accuracy [17].

V. Current Research Trends

Many efforts have taken place to overcome MI–BCI challenges. To the best of our knowledge, there are at least two distinguishable streams of research around the phenomenon; 1) Improving the machine part like signal acquisition, signal preprocessing, feature extraction, or classification. The other stream is improving the human factor. It includes user training, neurofeedback, or research on experimental paradigm.

A. Adaptive Decoding in MI-BCI

Adaptive decoding algorithms dynamically update classifier parameters to address EEG drift and changing conditions [2], [11]Two main strategies are prevalent. They depend primarily on retrain and rebiase concepts [45]. Supervised adaptation which incorporates true labels after classification to refine decision boundaries can be highly effective but requires labeled data in real time [2], [42]. On the other hand, unsupervised adaptation relies on predicted labels, adjusting parameters without ground truth [11] which may result in potential drift in predictions, it alleviates the need for explicit user feedback. Reinforcement signals, like error–related potentials (ErrPs) can facilitate adaptive processes by indicating classification mistakes [76].

B. Transfer Learning in MI-BCI

Transfer learning (TL) addresses data insufficiency and variability by repurposing knowledge from existing domains, datasets, or models [16], [77]. The transfer can be applied on subject-, session-, dataset-, or domain-levels. In Subject–to–Subject Transfer, models are trained on large cohorts to bootstrap new user sessions [78]. Session–to–session transfer reduces repeated calibration within the same user [79]. Domain-to-domain transfer can be considered domain adaptation when the knowledge is transferred to a seen domain. On the other hand, if the domain is unseen, the TL approach is considered Domain Generalization [14], [80]. With TL, MI–BCI systems can shorten training times and maintain performance even for users with initially weak SMRs [21].

C. User Training and Feedback Mechanisms

Given the significant influence of human factors in MI–BCI, research increasingly focuses on refined training protocols. User-focused interventions, including training protocols and enhanced feedback, are increasingly recognized as vital to improving MI–BCI performance, particularly for individuals labeled as BCI–illiterate.

Real-time neurofeedback—visual, auditory, or haptic—can inform users when their MI strategy produces the desired cortical patterns [81], [82]. Individualized training considers user preferences, cognitive styles, or motivational levels, enhancing skill acquisition and potentially reducing BCI–illiteracy rates [48].

Some researchers argue that anatomical differences may limit detectable SMRs in certain subjects, yet other studies emphasize shortcomings in the standard MI–BCI training protocol [48], [83]. In a 54–subject experiment, Jeunet et al. [48] reported that about 17% could not master basic motor tasks, highlighting the urgency for more effective training strategies. These results suggest that targeted instruction, grounded in techniques from educational psychology and instructional design, could bolster MI–BCI performance, although much remains to be tested in real-world scenarios [83]. Additionally, factors such as user motivation, visuospatial memory, and vividness of visual imagery—identified by Leeuwis et al. [84]—imply that addressing individualized cognitive traits may further optimize user training.

Researchers have also proposed refining experimental paradigms by integrating feedback or error correction mechanisms to assist users during MI–BCI sessions [12], [49]. The rationale is that immediate, adaptive feedback can help subjects fine-tune their mental strategies in real time, fostering stronger and more consistent SMRs. For example, a system might visually indicate the quality of the user's ongoing motor imagery, allowing them to adjust their cognitive approach and reduce reliance on guesswork. Although these paradigm enhancements remain under study, they constitute part of a broader push to move beyond mere machine-centric optimizations. By combining robust feature extraction and classification methods with user-friendly training approaches, researchers hope to lower BI rates and ensure a larger fraction of participants can achieve reliable MI–BCI control.

D. Deep Learning in MI-BCI

Deep learning has emerged as a valuable approach in MI–BCI research, helping address a variety of challenges and enhancing overall system performance including signal preprocessing, feature extraction, and classification. Recent works have explored deep learning approaches to enhance MI–BCI performance. Tibrewal et al. demonstrated that a fine-tuned CNN classifier could boost overall accuracy by 18.64% for insufficient BCI users compared to conventional methods [23]. In another effort, Kim et al. proposed a subject-to-subject semantic style transfer network, leveraging class-discriminative styles from users with good MI-BCI accuracy to improve decoding accuracy for others [21]. Their method surpassed competing approaches on MI classification tasks. Li et al. suggested that the low performance of MI-BCI is not purely physiological by introducing a multi-direction transfer learning (MDTL) strategy, enabling knowledge transfer from multiple source models and significantly improving accuracy for subjects with low performance using conventional deep learning algorithms [85]. Saadati et al. [35] incorporated DNNs in a hybrid EEG–fNIRS system, showing considerable classification accuracy across MI and mental workload tasks.

Earlier research also integrated deep learning with established feature extraction methods to better capture spatiotemporal EEG patterns. For instance, Stacked Denoising Autoencoders (SDA) have been combined with Common Spatial Patterns (CSP) to extract higher-level features, improving multi-class MI–BCI classification [34]. In a related strategy, Abbasi et al. [27] merged continuous wavelet transforms (CWT) with a deep learning architecture based on AlexNet, demonstrating promising accuracy gains. Another effort, Salimpour et al. [78], utilized transfer learning from both AlexNet and VGG19 by converting EEG signals into image-like inputs, easing calibration requirements and enhancing classification efficiency. While these techniques offer promising avenues for addressing challenges such as insufficient MI-BCI users, they often require extensive calibration data or careful selection of parameters, indicating that further refinements, particularly in profiling and accommodating insufficient subjects, remain a key area of ongoing research. .

Beyond these hybrid pipelines, researchers have explored specialized deep learning architectures that capture complex spatiotemporal EEG dynamics. Spiking neural networks (SNNs) were present in the literature. Tang et al. [86] used SNNs in conjunction with FBCSP and F-score optimization, noting moderate performance for MI classification. Three datasets have been used to validate a spiking neural network which is adaptive (SCNet) [87]. Although the network outperformed the state of the art in 2–class setting, its performance in 4-class decoding (67.43%) was below the accepted accuracy for reliable BCI (70%).

Graph convolutional neural networks (GCNs) have presence in the literature as well. They can leverage brain functional connectivity metrics like imaginary coherence (iCOH). Bian et. al. [88] used GCNs and Temporal Convolutional Networks (TCNs) with multi-head attention to integrate spatial and temporal features. An adaptive GCN were used in decoding MI trials with multi–branch graph adaptive network (MBGA–Net). It adapts by assigning relevant time–frequency domain using augmented EEG signal [89]. Multi-head attention mechanism is used in [36] with spatio-temporal convolution in a hybrid approach.

Another line involves focusing on attention-enhanced CNNs. Double Attention ConvNet (DAttnConvNet) achieved high accuracy in EEG decoding [90]. Li et al. [91] combined a novel artifact removal strategy with a Spatial Attention-Based Multiscale CNN (SA-MSCNN), reaching 79.83% average accuracy on the BCI Competition IV dataset 2a. In parallel, L. Wang & Li [92] explored Dipole Feature Imaging (DFI), converting 3D dipole information into 2D feature images processed by hybrid CNNs to capture MI features.

Recent work has also targeted BCI–illiteracy (BI) and insufficient BCI accuracy using deep learning strategies. Tao et al. [47] proposed a self-adaptive multiple kernel extreme learning machine framework aimed at insufficient subjects, though it still averaged below 58% accuracy. In contrast, Tibrewal et al. [23] introduced a fine-tuned CNN that provided an 18.64% boost for insufficient BCI users relative to conventional models, without profiling BI. A subject-to-subject semantic style transfer network, drawing on class-discriminative patterns from individuals less affected by BCI-illiteracy was proposed by [21]. Multi-direction transfer learning (MDTL) approach reported a significant accuracy gains for BCI-illiterate subjects with conventional deep learning algorithms.

Altogether, these studies illustrate how deep learning methods—ranging from CNN variants and autoencoders to spiking networks and attention-based models—offer diverse paths to refining MI–BCI classification. By incorporating specialized feature representations, employing transfer learning to reduce calibration, or targeting BI-specific user groups, researchers continue to expand the scope and impact of deep learning in MI–BCI. The varied architectures and training protocols underscore the complexity of decoding EEG signals, while highlighting deep learning's promise in addressing traditional MI–BCI challenges.

VI. Future Directions

Many advancements can be future-seen in the literature. Those developments are natural evolvement of current trends. real-time and high-accuracy MI–BCIs, personalization of MI-BCI, usability, and commercial integration.

Improved dynamic artifact removal proposed by Tsai et al. [93] proposed a technique that dynamically adjust the artifact removal process. Their Adaptive Online Artifact Subspace Reconstruction (ASR) integrates Hebbian/anti-Hebbian neural networks into their algorithm to make it dynamic. The adaptive artifact removal was used in [94]. The authors suggested using Common Spatial Pattern (CSP) and Chaotic Particle Swarm Optimization (CPSO) to make a self-adaptive artifact removal and CSP to obtain distinguishable features, improving recognition accuracy significantly. Hybrid modalities like combining EEG with functional near-infrared spectroscopy (fNIRS), or merging sensory feedback with motor imagery, can enrich the overall signal space, improving confidence in user intent [35].

BCI–illiteracy remains a major bottleneck. Future work may delve deeper into neuroanatomical predictors and brain-state feedback to guide tailored interventions [20], [95]. Coupled with adaptive decoding, such individualized approaches could identify user-specific neural patterns, mitigate differences in SMR generation, and streamline training. Transfer learning and adaptive algorithms remain on the trend.

Creating robust classifiers that require minimal data from subjects for calibration is a major direction as well. Federated and collaborative learning may help making the calibration of classification model less data intensive. Federated learning triggers data privacy concerns where models update from multiple labs or user devices without centralizing raw EEG signals. For MI–BCI to transition from specialized labs to mainstream use and daily applications further refinements could be required. MI-BCI should be more efficient, convenient, and user-friendly. MI-BCI fatigue-free with reasonable transfer rate and less calibration requirement is a demand.

Besides the privacy concerns, other ethical considerations arise as well. EEG signal could potentially reveal sensitive cognitive states or personal identification necessitating careful regulations and technical solutions like encryption.

VII. Conclusion

Motor imagery–based BCIs embody a dynamic intersection of neuroscience, signal processing, and computational intelligence. The thematic analysis presented here highlights pivotal aspects found in recent literature. We, first, exhibited the fundamentals of MI–BCI. Then, we elaborated on core technical modules. Next, we highlighted the challenges and limitations such as BCI–illiteracy, data scarcity, and inter-subject variability. After than, we investigated the latest research direction found in the surveyed literature. Finally, we anticipated future research directions.

Researchers, engineers, clinicians, and end-users must work in collaboration to refine training protocols, scale adaptive algorithms, and innovate hardware design. As the field moves beyond proof-of-concept demonstrations toward integrated BCI solutions, MI–BCIs stand poised to transform communication, rehabilitation, and human–machine interaction on an unprecedented scale. Ensuring that these systems are equitable, reliable, and accessible.

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